

OpenStax Calculus Volume 3

Section 3.3 Exercises: Arc Length and Curvature

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Find the arc length of the curve on the given interval.

1. $\mathbf{r}(t) = t^2\mathbf{i} + 14t\mathbf{j}$, $0 \leq t \leq 7$. This portion of the graph is shown here:
[Figure: This figure is the graph of a curve beginning at the origin and increasing.]
2. $\mathbf{r}(t) = t^2\mathbf{i} + (2t^2 + 1)\mathbf{j}$, $1 \leq t \leq 3$
3. $\mathbf{r}(t) = \langle 2 \sin t, 5t, 2 \cos t \rangle$, $0 \leq t \leq \pi$. This portion of the graph is shown here:
[Figure: This figure is the graph of a curve in 3 dimensions. It is inside of a box. The box represents an octant. The curve begins in the upper right corner of the box and bends through the box to the other side.]
4. $\mathbf{r}(t) = \langle t^2 + 1, 4t^3 + 3 \rangle$, $-1 \leq t \leq 0$
5. $\mathbf{r}(t) = \langle e^{-t} \cos t, e^{-t} \sin t \rangle$ over the interval $[0, \frac{\pi}{2}]$. Here is the portion of the graph on the indicated interval:
[Figure: This figure is the graph of a curve in the first quadrant. It begins approximately at 0.20 on the y-axis and increases to approximately where $x = 0.3$. Then the curve decreases, meeting the x-axis at 1.0.]
6. Find the length of one turn of the helix given by $\mathbf{r}(t) = \frac{1}{2} \cos t \mathbf{i} + \frac{1}{2} \sin t \mathbf{j} + \sqrt{\frac{3}{4}} t \mathbf{k}$.
7. Find the arc length of the vector-valued function $\mathbf{r}(t) = -t\mathbf{i} + 4t\mathbf{j} + 3t\mathbf{k}$ over $[0, 1]$.
8. A particle travels once around a circle with the equation of motion $\mathbf{r}(t) = 3 \cos t \mathbf{i} + 3 \sin t \mathbf{j} + 0\mathbf{k}$. Find the distance traveled around the circle by the particle.
9. Set up an integral to find the circumference of the ellipse with the equation $\mathbf{r}(t) = \cos t \mathbf{i} + 2 \sin t \mathbf{j} + 0\mathbf{k}$.
10. Find the length of the curve $\mathbf{r}(t) = \langle \sqrt{2}t, e^t, e^{-t} \rangle$ over the interval $0 \leq t \leq 1$. The graph is shown here:
[Figure: This figure is the graph of a curve in 3 dimensions. It is inside of a box. The box represents an octant. The curve begins in the upper left corner of the box and bends through the box to the bottom of the other side.]
11. Find the length of the curve $\mathbf{r}(t) = \langle 2 \sin t, 5t, 2 \cos t \rangle$ for $t \in [-10, 10]$.
12. The position function for a particle is $\mathbf{r}(t) = a \cos(\omega t) \mathbf{i} + b \sin(\omega t) \mathbf{j}$. Find the unit tangent vector and the unit normal vector at $t = 0$.

13. Given $\mathbf{r}(t) = a \cos(\omega t)\mathbf{i} + b \sin(\omega t)\mathbf{j}$, find the binormal vector $\mathbf{B}(0)$.
14. Given $\mathbf{r}(t) = \langle 2e^t, e^t \cos t, e^t \sin t \rangle$, determine the tangent vector $\mathbf{T}(t)$.
15. Given $\mathbf{r}(t) = \langle 2e^t, e^t \cos t, e^t \sin t \rangle$, determine the unit tangent vector $\mathbf{T}(t)$ evaluated at $t = 0$.
16. Given $\mathbf{r}(t) = \langle 2e^t, e^t \cos t, e^t \sin t \rangle$, find the unit normal vector $\mathbf{N}(t)$ evaluated at $t = 0$, $\mathbf{N}(0)$.
17. Given $\mathbf{r}(t) = \langle 2e^t, e^t \cos t, e^t \sin t \rangle$, find the unit binormal vector evaluated at $t = 0$.
18. Given $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t\mathbf{k}$, find the unit tangent vector $\mathbf{T}(t)$. The graph is shown here:
[Figure: This figure is the graph of a curve in 3 dimensions. It is inside of a box. The box represents an octant. The curve begins in the bottom left corner of the box and bends through the box to the upper left side.]
19. Find the unit tangent vector $\mathbf{T}(t)$ and unit normal vector $\mathbf{N}(t)$ at $t = 0$ for the plane curve $\mathbf{r}(t) = \langle t^3 - 4t, 5t^2 - 2 \rangle$. The graph is shown here:
[Figure: This figure is the graph of a curve above the x-axis. The curve decreases in the second quadrant, passes through the y-axis at $y = 20$. Then it intersects the origin. The curve loops at the origin, increasing back through $y = 20$ into the first quadrant.]
20. Find the unit tangent vector $\mathbf{T}(t)$ for $\mathbf{r}(t) = 3t\mathbf{i} + 5t^2\mathbf{j} + 2t\mathbf{k}$
21. Find the principal normal vector to the curve $\mathbf{r}(t) = \langle 6 \cos t, 6 \sin t \rangle$ at the point determined by $t = \pi/3$.
22. Find $\mathbf{T}(t)$ for the curve $\mathbf{r}(t) = (t^3 - 4t)\mathbf{i} + (5t^2 - 2)\mathbf{j}$.
23. Find $\mathbf{N}(t)$ for the curve $\mathbf{r}(t) = (t^3 - 4t)\mathbf{i} + (5t^2 - 2)\mathbf{j}$.
24. Find the unit normal vector $\mathbf{N}(t)$ for $\mathbf{r}(t) = \langle 2 \sin t, 5t, 2 \cos t \rangle$.
25. Find the unit tangent vector $\mathbf{T}(t)$ for $\mathbf{r}(t) = \langle 2 \sin t, 5t, 2 \cos t \rangle$.
26. Find the arc-length function $s(t)$ for the line segment given by $\mathbf{r}(t) = \langle 3 - 3t, 4t \rangle$. Write $\mathbf{r}$ as a parameter of s .
27. Parameterize the helix $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$ using the arc-length parameter s , from $t = 0$.
28. Parameterize the curve using the arc-length parameter s , at the point at which $t = 0$ for $\mathbf{r}(t) = e^t \sin t\mathbf{i} + e^t \cos t\mathbf{j}$.
29. Find the curvature of the curve $\mathbf{r}(t) = 5 \cos t\mathbf{i} + 4 \sin t\mathbf{j}$ at $t = \pi/3$. (Note: The graph is an ellipse.)
[Figure: This figure is the graph of an ellipse. The ellipse is oval along the x-axis. It is centered at the origin and intersects the y-axis at -4 and 4 .]
30. Find the x -coordinate at which the curvature of the curve $y = 1/x$ is a maximum value.
31. Find the curvature of the curve $\mathbf{r}(t) = 5 \cos t\mathbf{i} + 5 \sin t\mathbf{j}$. Does the curvature depend upon the parameter t ?
32. Find the curvature κ for the curve $y = x - \frac{1}{4}x^2$ at the point $x = 2$.
33. Find the curvature κ for the curve $y = \frac{1}{3}x^3$ at the point $x = 1$.
34. Find the curvature κ of the curve $\mathbf{r}(t) = t\mathbf{i} + 6t^2\mathbf{j} + 4t\mathbf{k}$. The graph is shown here:
[Figure: This figure is the graph of a curve in 3 dimensions. It is inside of a box. The box represents an octant. The curve has a parabolic shape in the middle of the box.]

35. Find the curvature of $\mathbf{r}(t) = \langle 2 \sin t, 5t, 2 \cos t \rangle$.
36. Find the curvature of $\mathbf{r}(t) = \sqrt{2}t\mathbf{i} + e^t\mathbf{j} + e^{-t}\mathbf{k}$ at point $P(0, 1, 1)$.
37. At what point does the curve $y = e^x$ have maximum curvature?
38. What happens to the curvature as $x \rightarrow \infty$ for the curve $y = e^x$?
39. Find the point of maximum curvature on the curve $y = \ln x$.
40. Find the equations of the normal plane and the osculating plane of the curve $\mathbf{r}(t) = \langle 2 \sin(3t), t, 2 \cos(3t) \rangle$ at point $(0, \pi, -2)$.
41. Find equations of the osculating circles of the ellipse $4y^2 + 9x^2 = 36$ at the points $(2, 0)$ and $(0, 3)$.
42. Find the equation for the osculating plane at point $t = \pi/4$ on the curve $\mathbf{r}(t) = \cos(2t)\mathbf{i} + \sin(2t)\mathbf{j} + t\mathbf{k}$.
43. Find the radius of curvature of $6y = x^3$ at the point $(2, \frac{4}{3})$.
44. Find the curvature at each point (x, y) on the hyperbola $\mathbf{r}(t) = \langle a \cosh(t), b \sinh(t) \rangle$.
45. Calculate the curvature of the circular helix $\mathbf{r}(t) = r \sin(t)\mathbf{i} + r \cos(t)\mathbf{j} + t\mathbf{k}$.
46. Find the radius of curvature of $y = \ln(x + 1)$ at point $(2, \ln 3)$.
47. Find the radius of curvature of the hyperbola $xy = 1$ at point $(1, 1)$.

A particle moves along the plane curve C described by $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j}$. Solve the following problems.

48. Find the length of the curve over the interval $[0, 2]$.
49. Find the curvature of the plane curve at $t = 0, 1, 2$.
50. Describe the curvature as t increases from $t = 0$ to $t = 2$.

The surface of a large cup is formed by revolving the graph of the function $y = 0.25x^{1.6}$ from $x = 0$ to $x = 5$ about the y -axis (measured in centimeters).

51. [T] Use technology to graph the surface.
52. Find the curvature κ of the generating curve as a function of x .
53. [T] Use technology to graph the curvature function.